

Calibrated Uncertainty and Selective Classification in Financial Fraud Detection

Project Proposal

Group: Intellicore

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1. Background and Introduction

The digitalization of global finance has exponentially increased the volume and velocity of transactions, bringing with it a corresponding rise in financial crimes such as credit card fraud and money laundering. To combat this, financial institutions heavily rely on Machine Learning classification models to monitor transactions in real time.

Despite their high accuracy, automated fraud detection is a strictly safety-critical domain. The cost of a false negative can result in millions of dollars in stolen funds and severe regulatory penalties. Conversely, the cost of a false positive results in frozen accounts, ruined customer experiences, and overwhelmed compliance teams.

In this high-stakes environment, simply outputting a binary prediction (Fraud vs. Genuine) is insufficient. Fraudsters continuously adapt their strategies, creating distribution shifts. When ML models encounter these novel, out-of-distribution tactics, they must not only be highly accurate but also well-calibrated, meaning their output probabilities must accurately reflect the true likelihood of correctness. Furthermore, to balance automation with safety, these systems require an integrated "ML with rejection" mechanism. By quantifying predictive uncertainty, the model can automatically approve or block clear-cut cases, while purposefully abstaining from ambiguous transactions and deferring them to human compliance officers for manual review.

2. Literature Review

Fraud-detection research has recently optimised heavily for discriminative performance while paying comparatively little attention to whether a model's probabilities can be trusted or whether it recognises when to stop. Naqvi's [1] systematic review of work from 2020 to 2025, which screened more than a thousand records and examined seventy-seven studies in depth, makes this explicit: it argues that models should be assessed along an operational-capability axis covering data realism, robustness to drifting fraud patterns, and governance for human investigators, and it reports that probability calibration and the setting of operational decision thresholds are routinely absent despite being essential to real workflows. The review concludes that detectors ought to be paired with explicit decision policies, including the routing of uncertain cases to analysts.

A first line of work responds by attaching uncertainty estimates to predictions rather than reporting bare labels. Habibpour et al. [2] are representative: observing that deep networks for card fraud have been tuned for point accuracy while their confidence goes unmeasured, they apply Bayesian-style estimators, Monte Carlo Dropout, and deep ensembles, to quantify predictive uncertainty, and show that the resulting estimates surface borderline cases and let a system abstain when confidence is low. Their motivation is precisely the safety-critical framing adopted here: because fraudsters change tactics continually, the model repeatedly meets data unlike its training distribution, and because expert review is slow, only a small fraction of transactions are ever verified. The practical drawback is cost, since these sampling-based estimators require many forward passes per transaction, awkward under the millisecond budgets of real-time screening.

A second, more foundational line concerns calibration itself. Guo et al. [3] demonstrated that modern deep networks, unlike their smaller predecessors, are systematically overconfident: their reported probabilities exceed their empirical accuracy, an effect tied to architectural factors such as depth, width, and batch normalisation. Their proposed remedy, temperature scaling, is a single-parameter post-hoc rescaling of the logits that leaves the predicted class unchanged but markedly improves calibration, and the Expected Calibration Error (ECE) they popularised has become the standard diagnostic. The result is central to any rejection scheme, because a policy that abstains or acts on the basis of a probability threshold is only as reliable as the calibration of the scores beneath it.

A third line moves from estimating uncertainty to acting on it inside a human-in-the-loop workflow. The closest recent example is the Integrated Transparency and Confidence Framework of Mapaila and Senekane [4], which combines local explanations (LIME) with split conformal prediction to produce calibrated prediction sets with formal coverage guarantees, evaluated on the large PaySim mobile-money dataset using Random Forest and XGBoost. When the framework cannot assign a label confidently, signalled by uncertain or empty prediction regions, it defers the case to an analyst, together with an explanation to guide the review. This is close in spirit to what we propose, yet it treats uncertainty as a post-hoc wrapper around tree-ensemble classifiers rather than a quantity the model produces intrinsically, and it does not separate uncertainty arising from genuine novelty from uncertainty arising from noisy data.

Underlying all of these efforts is the problem of data. Realistic transaction records are scarce because of privacy and legal constraints, and the public datasets that do exist tend to lack diversity and reliable labels. Oztas et al. [5] address this by releasing SAML-D, a synthetic anti-money-laundering dataset generated from a purpose-built transaction simulator, comprising twelve features and twenty-eight normal and suspicious typologies across a range of geographies and payment types; tellingly, standard models perform worse on it than on older benchmarks, which makes it a demanding testbed for behaviour under rare and deceptive patterns exactly the regime in which a model ought to abstain. Taken together, the literature assembles the necessary ingredients for uncertainty estimation, calibration, deferral, and realistic evaluation, but largely treats them in isolation: uncertainty methods are often expensive and themselves uncalibrated, calibration work rarely connects to a rejection policy, and the one framework that ties rejection to human review does so outside the model and without disentangling the kind of uncertainty involved. Closing that fragmentation is the aim of this proposal.

3. Problem Statement

Most fraud-detection pipelines are optimised almost entirely for classification metrics such as AUPRC or F1, while predictive uncertainty and calibration receive little attention. Three connected shortcomings follow. First, the probabilities are not trustworthy as probabilities: networks trained with cross-entropy are overconfident [3], so a transaction with novel characteristics can still receive a 99% score, and such poorly calibrated outputs (high ECE) are an unreliable basis for automated thresholds.

Second, most systems offer no principled way to abstain. They force a binary verdict on every transaction, so the model must guess even with no relevant experience – a direct contributor to high false-positive rates. What is missing is the ability to separate epistemic uncertainty (ignorance when meeting OOD tactics) from aleatoric uncertainty (noisy or ambiguous data) and to defer on the former.

Third, the approaches that do quantify uncertainty are not yet a comfortable fit for real-time screening: the sampling methods of Habibpour et al. [2] need many forward passes, while the conformal wrapper of Mapaila and Senekane [4], though well suited to deferral, sits on base scores whose calibration is not guaranteed and does not disentangle the source of uncertainty. Naqvi's [1] review confirms field-wide that calibration and explicit rejection policies remain the exception. The problem we target is therefore the absence of a single, deployable model that produces calibrated probabilities, distinguishes epistemic from aleatoric uncertainty in one forward pass, and uses that uncertainty to decide when to defer evaluation under realistic distribution shift.

4. Proposed Method

We propose a framework combining Evidential Deep Learning (EDL) for single-pass uncertainty estimation, post-hoc calibration of the confidence scores, and a selective classification policy built on the estimated uncertainty. The three components target the three shortcomings above and may be refined as the project progresses.

An uncertainty quantification with EDL. Rather than the multiple passes that Deep Ensembles or MC Dropout require, we formulate the classifier using EDL [6]. The network outputs the parameters of a Dirichlet distribution over class probabilities, treating predictions as evidence-based opinions. From a single forward pass, this yields an aleatoric component and an epistemic component. The single-pass property is what makes the approach attractive under latency constraints.

Post-hoc calibration. To make the scores usable for threshold-based decisions, we calibrate them after training, primarily with temperature scaling [3], with calibration-aware losses such as focal loss as an alternative. Calibration quality is measured with ECE and the Brier score. Because fraud data is extremely imbalanced, the mass of legitimate transactions can mask poor calibration on rare fraud cases, so calibration is examined on the minority class specifically, not only in aggregate.

Selective classification. We then attach a rejection policy. Let $U(x)$ be the normalised epistemic uncertainty for transaction x and $\tau \in [0, 1]$ an operating threshold reflecting risk tolerance. If $U(x) \leq \tau$, the system acts automatically; if $U(x) > \tau$, it abstains and routes the case, with supporting information, to a human officer. Sweeping τ traces the trade-off between coverage and reliability, reported via accuracy rejection and risk coverage curves. Baselines include softmax with temperature scaling, MC Dropout, and Deep Ensembles, so the contribution of each component can be assessed separately.

5. Available Datasets

Three public datasets cover the conditions we need to test calibration, OOD behaviour, and the reject option.

- **Credit Card Fraud Detection (ULB, 2013).** Real European card transactions over two days in September 2013: 284,807 transactions, 492 fraudulent, with PCA-transformed features plus Time and Amount. Its extreme imbalance makes it a natural baseline for ECE and post-hoc calibration. kaggle.com/datasets/mlg-ulb/creditcardfraud.
- **PaySim.** A synthetic mobile-money simulator built from a one-month sample of real African transactions with raw, interpretable features, well suited to OOD detection and rejection on multi-step fraud, and the dataset used by Mapaila and Senekane [4]. kaggle.com/datasets/ealaxi/paysim1.
- **SAML-D.** The synthetic AML dataset of Oztas et al. [5]: 12 features, 28 normal and suspicious typologies, and a very low suspicious rate. It's rare typologies suit the learning to defer setting, common patterns handled automatically, rarer ones deferred under high uncertainty. kaggle.com/datasets/berkanoztas/synthetic-transaction-monitoring-dataset-aml.

References

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